MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG2DSCIAM100 - Intelligent Automation Techniques

Duration: 2 hour **Maximum Marks: 35**

PracticalExamination–ModelQuestionPaper

Instructions:

- **Demonstrate**thegiventask
- **Viva session** (Minimum 5 questions from the PracticalModule)
- **LabReport**

EvaluationCriteria:

- **Viva**:7 marks
- **LabReport**:8marks
- **Demonstration**:20marks

Tasks:*(Chooseanyone)*

1. Using past temperature and humidity data, develop a simple algorithm to predicttomorrow's weather conditions (sunny, rainy, or cloudy). Implement your solution using Python and display the predicted outcome.
2. (a) Write down the steps involved in predicting the house price using Linear Regression
   (b) Implement the above procedure in Python using appropriate dataset.
3. (a) Write down the steps involved for Handwritten digit recognition using KNN
   (b) Implement the above procedure in Python using appropriate dataset.
4. (a) Write down the steps involved for Handwritten digit recognition using Gradient descent
   (b) Implement the above procedure in Python using appropriate dataset.
5. (a) Write down the steps involved in predicting disease using Linear Regression.
   (b) Implement the above procedure in Python using appropriate dataset.
6. (a) Write a Python program using KNN to classify different types of fruits based on features such as colour, size, and weight.
7. Implement KNN regression to predict the house prices from a dataset.